Live State Estimation and Air Brake Controls FDR

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High Level Overview

System Overview

- Apogee prediction will accurately predict final apogee from sensor readout
- Controls will adjust drag based on apogee predicted to change state to reach predefined target apogee

Requirements

Index	Description	Comments
LSE1	Estimation should completed in time less than time step	Look-up tables will be effective here
LSE2	Must handle noise effectively such that covariance converges	Tune filter and run simulations/L3 runs
LSE3	Ensure that system does not force the airbrakes to do unfeasible things	Install “clamps” that, if the reaching target apogee is physically not possible, deploy max and hope for the best
LSE4	Estimator must recover from large initial uncertainty quickly	Have reasonable error bounds and depend on physics model if necessary
LSE5	Estimator must be robust to intermittent sensor dropouts	Make robust physics model and fine tune coeffiecents

System Integration

- Will integrate directly to mechanical air-brakes system.
- Code will go through CFC to send information air-brakes on how much to open/retract

Sensors

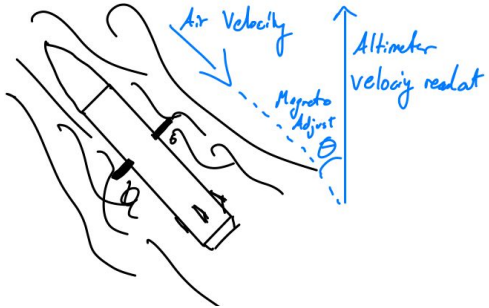
Sensor Details

Magnetometer AK09919C Family	Gyroscope (IMU) ICM-42688-P	Barometer BMP390
Operating Range: –30 °C → +85 °C Refresh Rate: 100 Hz Accuracy Error (Magnetic Noise): Low-Noise Mode: 0.15–0.25 μT → 2–4 ms delay Normal Mode: 0.6–0.8 μT ~1-3°	Measurement Range: ±250, ±500, ±1000, ±2000 °/s Refresh Rate (ODR): 8,000 Hz Bias: ±0.5 °/s (typical) ±3 °/s (max) Sensitivity Error: ±0.5% typical RMS Noise: ≈ 0.002 °/s/√Hz (typical) At 1 kHz filtered data: ~0.06–0.08 °/s RMS noise	Pressure Range: 300 – 1100 hPa Refresh Rate: 200 Hz Accuracy: Absolute: ±0.50 hPa → ≈ ±0.5 – 0.7 m height error Relative: ±0.03 hPa (T = 25 – 40 °C) RMS Noise: 0.02 Pa (at lowest bandwidth, highest resolution)

Sensor Discussion

Magnetometer & Gyroscope

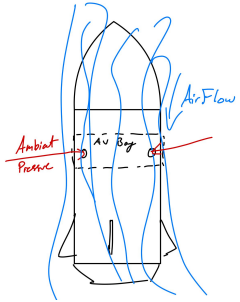
Need to know angle to find air speed to calculate drag in direction of flow



Angle will allow us to calculate drag force from vertical velocity

Barometer BMP390

Altitude calculated from delta of pressure from ground



Interesting issue in measuring air pressure, vertical airflow can act like a wall and cause delay in sensor read out (discussed more in next slide)

Error is so low, within bounds, and non-cumulative, so we do not need filter

Two-Types of Delays

Navigating Altitude Delays

GPS altitude is not affected by airspeed and the time to equilibrate with the surrounding air pressure. It is undecided if barometer reads delay or ahead. Although GPS data has large error, L3 launch data will help learn about how the barometer works and the truth in the altitude it reads out.

2 Methods

Static Adjustment

If L3 launch data suggests a linear delay in barometer readout vs GPS, we can scale the barometer readout by a fixed value after launch

Live Adjustment

If L3 launch data suggests non linear delay in barometer readout vs GPS, we scale the barometer readout everytime a GPS output is printed and use that scale factor for that time step

Navigating Deployment Delays

Software

Get ack from motor saying that operation is completed and do not send further operations until current is done.

When current operation is done continue in LIFO order because sensor data is the most relevant since there are external factors affecting the rocket flight

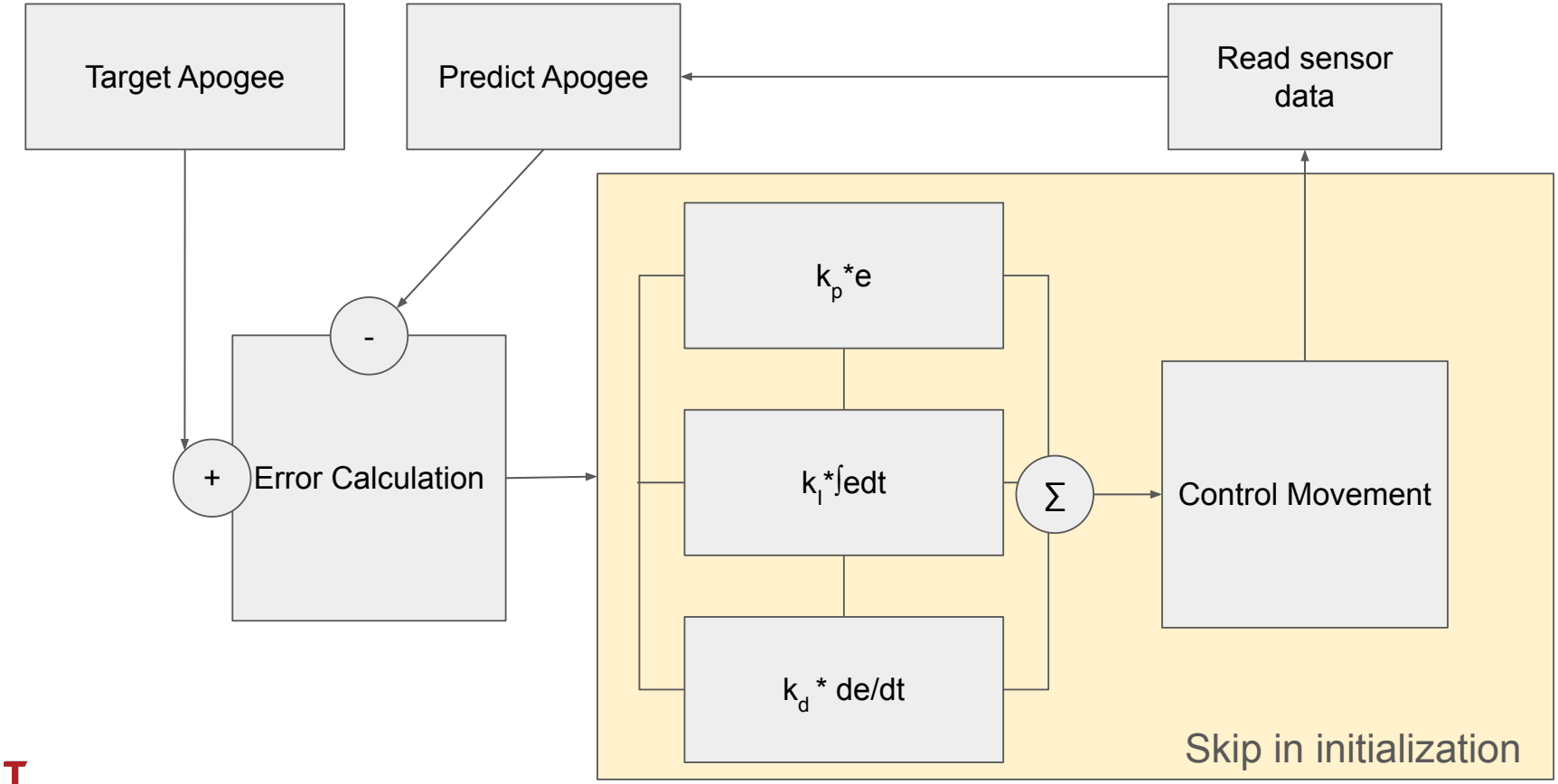
Controller

Adjust coefficient of P and I (k_p, k_i) in PID controller to adjust level of adjustment.

This can prevent the controller from making the air brakes fully deploy and fully retract constantly

Controls

Control Flow



Apogee Prediction

Numerical solver that will simulate flight starting with current states

Input States:

Position
Velocity
Gravity
Degrees off trajectory
Air Resistance

→

Iterate through states and uses kinematics with timestep ~0.1s. To find next state vector until velocity = 0

→

Expected apogee based on current states

Let's Walk Through Two Time Steps

t = 6.9	t = 7 (time of GPS update)
Receive data on height, velocity, acceleration, drag, tilt	Re-read data on height, velocity, acceleration, (adjust by GPS error), drag, tilt from sensors
Run apogee prediction based on these variables	Run apogee prediction based on these variables
Calculate error and output to PID controller	Calculate error and output to PID controller
Deploy/retract brake accordingly	Deploy/retract brake accordingly

Continue until apogee is reached

Airbrake Deployment Algorithm

Challenge

- Comp requirement states that we must wait until thrust has ended, maximum aerodynamic pressure has been reached, or rocket has crossed 6,500 feet. Whichever comes first
- Should hold off as long as possible because once rocket slows down there is no speeding up
- Should be early enough that velocity is high enough to have a meaningful effect on speed

Strategy: Begin Passive and Become More Aggressive as Apogee Approaches

The moment after boost phase ends, the algorithm begins deploying the air brakes with the goal of reaching 10,000 feet being in the middle range of airbrake deployment. That is, with airbrakes fully deployed, we will undershoot, with airbrakes retracted, we will overshoot

Next, the algorithm looks to find the drag force needed to reach this target and makes this drag force constant by retracting in the beginning and expanding as time continues until apogee where it will be fully deployed

Computing

Computing

Arithmetic Operations	Special Operations
Adds/Subs 10 cycles: ~3,697 (37k cycles)	exp() 1,000 cycles : ~517 calls (517k cycles)
Mults 20 cycles: ~8,471(169k cycles)	Cholesky 4,000 cycles (7×7): 1
Divs 80 cycles: ~1,505 (120k cycles)	2×2 inverse 250 cycles: 1

Clock on 133 MHz Pico: 133 MHz => period = 1 / 133e6 ≈ 7.52 ns per cycle.

Time = 848,040 cycles × 7.52 ns ≈ 6.376 ms per control step << 0.01 time step

Progress Thus Far…

TARGET_APOGEE (ft)	9842.51968503937
GRAVITATIONAL_CONSTANT (m^3 kg^-1 s^-2)	6.6743e-11
MASS_OF_ROCKET (kg)	100
MASS_OF_EARTH (kg)	5.972e+24
SEA_LEVEL (ft)	20902230.971128605
STARTING_HEIGHT_AT_SITE (ft)	1640.4199475065616
TIME_STEP (s)	0.05
GROUND_TEMPERATURE (K)	300
Q scalar (process noise)	0.001
R pos variance (ft^2)	269.097760417743
R vel variance ((ft/s)^2)	43.05564166683888
P scalar (initial cov)	1.0

☑ Start from burnout initial conditions

Burnout height (ft)	5990.0
Burnout speed (ft/s)	731.0
Burnout acceleration (ft/s^2)	-32.174
Min drag (lbf)	30.0
Max drag (lbf)	100.0

Start Stop

```
t=14.60s | error(final apogee)=-2408.84 ft | u(Drag)=100.00 lbf | du(Drag)=130.83 lbf | h_true=12065.47 ft | h_est=12113.70 ft | apogee_pred=12251.36 ft
t=14.70s | error(final apogee)=-2400.39 ft | u(Drag)=100.00 lbf | du(Drag)=131.07 lbf | h_true=12073.02 ft | h_est=12119.76 ft | apogee_pred=12242.91 ft
t=14.80s | error(final apogee)=-2393.12 ft | u(Drag)=100.00 lbf | du(Drag)=130.59 lbf | h_true=12080.11 ft | h_est=12125.89 ft | apogee_pred=12235.64 ft
t=14.90s | error(final apogee)=-2388.34 ft | u(Drag)=100.00 lbf | du(Drag)=130.70 lbf | h_true=12086.75 ft | h_est=12132.34 ft | apogee_pred=12230.86 ft
t=15.00s | error(final apogee)=-2381.27 ft | u(Drag)=100.00 lbf | du(Drag)=129.54 lbf | h_true=12092.94 ft | h_est=12137.07 ft | apogee_pred=12223.79 ft
t=15.10s | error(final apogee)=-2375.98 ft | u(Drag)=100.00 lbf | du(Drag)=129.47 lbf | h_true=12098.60 ft | h_est=12142.31 ft | apogee_pred=12218.50 ft
t=15.20s | error(final apogee)=-2368.75 ft | u(Drag)=100.00 lbf | du(Drag)=129.10 lbf | h_true=12103.80 ft | h_est=12146.15 ft | apogee_pred=12211.27 ft
t=15.30s | error(final apogee)=-2364.64 ft | u(Drag)=100.00 lbf | du(Drag)=129.07 lbf | h_true=12108.57 ft | h_est=12150.55 ft | apogee_pred=12207.16 ft
t=15.40s | error(final apogee)=-2358.20 ft | u(Drag)=100.00 lbf | du(Drag)=128.47 lbf | h_true=12112.87 ft | h_est=12153.07 ft | apogee_pred=12200.72 ft
t=15.50s | error(final apogee)=-2353.92 ft | u(Drag)=100.00 lbf | du(Drag)=128.91 lbf | h_true=12116.72 ft | h_est=12156.33 ft | apogee_pred=12196.44 ft
t=15.60s | error(final apogee)=-2350.40 ft | u(Drag)=100.00 lbf | du(Drag)=128.14 lbf | h_true=12120.07 ft | h_est=12159.39 ft | apogee_pred=12192.92 ft
t=15.70s | error(final apogee)=-2343.96 ft | u(Drag)=100.00 lbf | du(Drag)=127.75 lbf | h_true=12122.90 ft | h_est=12160.20 ft | apogee_pred=12186.48 ft
t=15.80s | error(final apogee)=-2338.35 ft | u(Drag)=100.00 lbf | du(Drag)=127.87 lbf | h_true=12125.31 ft | h_est=12160.73 ft | apogee_pred=12180.87 ft
t=15.90s | error(final apogee)=-2332.33 ft | u(Drag)=100.00 lbf | du(Drag)=127.17 lbf | h_true=12127.23 ft | h_est=12160.26 ft | apogee_pred=12174.85 ft
t=16.00s | error(final apogee)=-2328.23 ft | u(Drag)=100.00 lbf | du(Drag)=127.46 lbf | h_true=12128.69 ft | h_est=12160.34 ft | apogee_pred=12170.75 ft
t=16.10s | error(final apogee)=-2324.40 ft | u(Drag)=100.00 lbf | du(Drag)=126.74 lbf | h_true=12129.65 ft | h_est=12160.11 ft | apogee_pred=12166.92 ft
t=16.20s | error(final apogee)=-2321.84 ft | u(Drag)=100.00 lbf | du(Drag)=126.92 lbf | h_true=12130.17 ft | h_est=12160.18 ft | apogee_pred=12164.36 ft
t=16.30s | error(final apogee)=-2319.79 ft | u(Drag)=100.00 lbf | du(Drag)=126.84 lbf | h_true=12130.23 ft | h_est=12159.92 ft | apogee_pred=12162.31 ft
Simulation complete.
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Max drag (lbf)	100.0

Start Stop

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t=14.90s | error(final apogee)=201.89 ft | u(Drag)=30.00 lbf | du(Drag)=-11.77 lbf | h_true=9507.20 ft | h_est=9545.23 ft | apogee_pred=9640.63 ft
t=15.00s | error(final apogee)=208.92 ft | u(Drag)=30.00 lbf | du(Drag)=-12.53 lbf | h_true=9513.20 ft | h_est=9549.52 ft | apogee_pred=9633.60 ft
t=15.10s | error(final apogee)=213.50 ft | u(Drag)=30.00 lbf | du(Drag)=-12.12 lbf | h_true=9518.85 ft | h_est=9554.22 ft | apogee_pred=9629.02 ft
t=15.20s | error(final apogee)=218.19 ft | u(Drag)=30.00 lbf | du(Drag)=-12.71 lbf | h_true=9524.14 ft | h_est=9558.47 ft | apogee_pred=9624.33 ft
t=15.30s | error(final apogee)=223.67 ft | u(Drag)=30.00 lbf | du(Drag)=-12.85 lbf | h_true=9529.06 ft | h_est=9561.75 ft | apogee_pred=9618.85 ft
t=15.40s | error(final apogee)=228.19 ft | u(Drag)=30.00 lbf | du(Drag)=-13.23 lbf | h_true=9533.56 ft | h_est=9565.01 ft | apogee_pred=9614.33 ft
t=15.50s | error(final apogee)=231.59 ft | u(Drag)=30.00 lbf | du(Drag)=-13.10 lbf | h_true=9537.77 ft | h_est=9568.37 ft | apogee_pred=9610.93 ft
t=15.60s | error(final apogee)=236.24 ft | u(Drag)=30.00 lbf | du(Drag)=-13.32 lbf | h_true=9541.55 ft | h_est=9570.59 ft | apogee_pred=9606.28 ft
t=15.70s | error(final apogee)=237.84 ft | u(Drag)=30.00 lbf | du(Drag)=-13.11 lbf | h_true=9544.98 ft | h_est=9574.16 ft | apogee_pred=9604.68 ft
t=15.80s | error(final apogee)=242.05 ft | u(Drag)=30.00 lbf | du(Drag)=-13.49 lbf | h_true=9548.03 ft | h_est=9575.66 ft | apogee_pred=9600.47 ft
t=15.90s | error(final apogee)=245.15 ft | u(Drag)=30.00 lbf | du(Drag)=-13.58 lbf | h_true=9550.74 ft | h_est=9577.48 ft | apogee_pred=9597.37 ft
t=16.00s | error(final apogee)=248.98 ft | u(Drag)=30.00 lbf | du(Drag)=-14.13 lbf | h_true=9553.06 ft | h_est=9578.51 ft | apogee_pred=9593.54 ft
t=16.10s | error(final apogee)=251.25 ft | u(Drag)=30.00 lbf | du(Drag)=-14.24 lbf | h_true=9555.01 ft | h_est=9579.78 ft | apogee_pred=9591.27 ft
t=16.20s | error(final apogee)=253.43 ft | u(Drag)=30.00 lbf | du(Drag)=-14.13 lbf | h_true=9556.63 ft | h_est=9580.80 ft | apogee_pred=9589.09 ft
t=16.30s | error(final apogee)=254.48 ft | u(Drag)=30.00 lbf | du(Drag)=-13.82 lbf | h_true=9557.84 ft | h_est=9582.05 ft | apogee_pred=9588.04 ft
t=16.40s | error(final apogee)=256.17 ft | u(Drag)=30.00 lbf | du(Drag)=-14.22 lbf | h_true=9558.71 ft | h_est=9582.51 ft | apogee_pred=9586.35 ft
t=16.50s | error(final apogee)=258.80 ft | u(Drag)=30.00 lbf | du(Drag)=-14.52 lbf | h_true=9559.21 ft | h_est=9581.59 ft | apogee_pred=9583.72 ft
t=16.60s | error(final apogee)=260.13 ft | u(Drag)=30.00 lbf | du(Drag)=-14.31 lbf | h_true=9559.33 ft | h_est=9581.42 ft | apogee_pred=9582.39 ft
Simulation complete.
```

☑ Start from burnout initial conditions

Burnout height (ft)	3990.0
Burnout speed (ft/s)	680.0
Burnout acceleration (ft/s^2)	-32.174
Min drag (lbf)	30.0
Max drag (lbf)	100.0

Start Stop

```
t=12.40s | error(final apogee)=20.52 ft | u(Drag)=79.20 lbf | du(Drag)=-1.62 lbf | h_true=9235.61 ft | h_est=9299.10 ft | apogee_pred=9822.00 ft
t=12.50s | error(final apogee)=26.76 ft | u(Drag)=75.17 lbf | du(Drag)=-1.75 lbf | h_true=9251.15 ft | h_est=9313.73 ft | apogee_pred=9815.76 ft
t=12.60s | error(final apogee)=33.31 ft | u(Drag)=69.69 lbf | du(Drag)=-2.83 lbf | h_true=9266.23 ft | h_est=9326.99 ft | apogee_pred=9809.21 ft
t=12.70s | error(final apogee)=37.65 ft | u(Drag)=64.36 lbf | du(Drag)=-2.84 lbf | h_true=9280.91 ft | h_est=9340.87 ft | apogee_pred=9804.87 ft
t=12.80s | error(final apogee)=38.65 ft | u(Drag)=59.53 lbf | du(Drag)=-2.42 lbf | h_true=9295.17 ft | h_est=9355.33 ft | apogee_pred=9803.87 ft
t=12.90s | error(final apogee)=43.69 ft | u(Drag)=54.38 lbf | du(Drag)=-3.06 lbf | h_true=9309.04 ft | h_est=9368.19 ft | apogee_pred=9798.83 ft
t=13.00s | error(final apogee)=49.10 ft | u(Drag)=47.87 lbf | du(Drag)=-3.52 lbf | h_true=9322.49 ft | h_est=9380.43 ft | apogee_pred=9793.42 ft
t=13.10s | error(final apogee)=51.29 ft | u(Drag)=41.61 lbf | du(Drag)=-3.02 lbf | h_true=9335.60 ft | h_est=9393.16 ft | apogee_pred=9791.23 ft
t=13.20s | error(final apogee)=53.07 ft | u(Drag)=34.81 lbf | du(Drag)=-3.66 lbf | h_true=9348.27 ft | h_est=9405.41 ft | apogee_pred=9789.45 ft
t=13.30s | error(final apogee)=57.17 ft | u(Drag)=30.00 lbf | du(Drag)=-3.26 lbf | h_true=9360.59 ft | h_est=9417.29 ft | apogee_pred=9785.35 ft
t=13.40s | error(final apogee)=70.02 ft | u(Drag)=30.00 lbf | du(Drag)=-4.89 lbf | h_true=9372.46 ft | h_est=9428.31 ft | apogee_pred=9772.50 ft
t=13.50s | error(final apogee)=79.70 ft | u(Drag)=30.00 lbf | du(Drag)=-5.55 lbf | h_true=9384.01 ft | h_est=9439.47 ft | apogee_pred=9762.82 ft
t=13.60s | error(final apogee)=91.32 ft | u(Drag)=30.00 lbf | du(Drag)=-6.48 lbf | h_true=9395.23 ft | h_est=9449.48 ft | apogee_pred=9751.20 ft
t=13.70s | error(final apogee)=99.48 ft | u(Drag)=30.00 lbf | du(Drag)=-6.59 lbf | h_true=9406.05 ft | h_est=9460.01 ft | apogee_pred=9743.04 ft
t=13.80s | error(final apogee)=112.34 ft | u(Drag)=30.00 lbf | du(Drag)=-7.74 lbf | h_true=9416.50 ft | h_est=9468.95 ft | apogee_pred=9730.18 ft
t=13.90s | error(final apogee)=124.95 ft | u(Drag)=30.00 lbf | du(Drag)=-8.49 lbf | h_true=9426.65 ft | h_est=9477.09 ft | apogee_pred=9717.56 ft
```

Next Steps

- Run through simulator and fine tune model
- Write up code in Rust
- Study L3 data to find data on GPS height vs barometer height
- Run magnetometer and gyro on L3 and study accuracy